

Nova SBE Econometrics

Class 12
Time Series: Further Issues

PAULO FAGANDINI

Fall 2025–26



Nova SBE Econometrics

Stationarity & Weak Dependence: A Refresher

Link to Slides



Strict Stationarity

Definition. A *strictly stationary* process is a stochastic process whose unconditional joint probability distribution *does not change when shifted in time*.

Stationary process

If you cut two slices of h periods apart you obtain similar moments of the distribution.

Nonstationary process

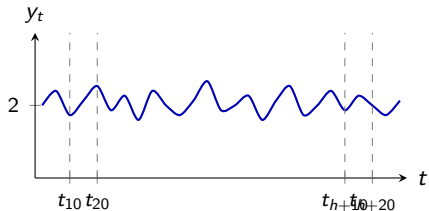
If you cut two slices of h periods apart you get different moments of the distribution.

Trending series

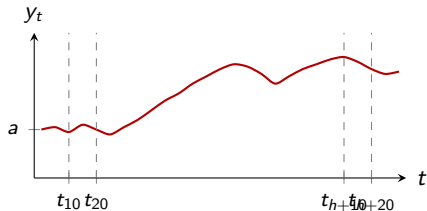
A trending series is **not** stationary — at least the mean changes over time. It can be **trend stationary**, i.e. stationary once the underlying time trend (a function solely of time) is removed.

Sample Paths: Stationary vs. Nonstationary

(A) Stationary process

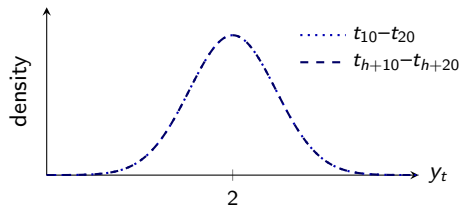


(B) Nonstationary process

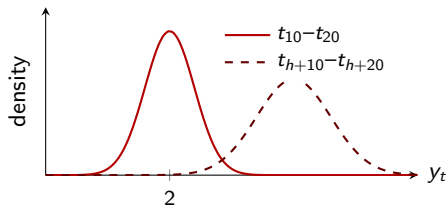


Densities of Two Slices (h Periods Apart)

(A) Stationary process



(B) Nonstationary process



Same shape (A) vs. shifted location/spread (B) illustrate why two slices behave so differently in the nonstationary case.

Why Do We Need Stationarity?

If strict exogeneity fails we need to rely on *weaker* assumptions. Intuitively, if the relationship between y and the regressors X is not stable over time, $\hat{\beta}_j$ cannot be pinned down — different “slices” of time deliver different results.

Covariance stationarity

A stochastic process $\{x_t : t = 1, 2, \dots\}$ with $E[x_t^2] < \infty$ is *covariance stationary* if:

- i. $E[x_t]$ is constant over time.
- ii. $V[x_t]$ is constant over time.
- iii. $\forall t, \forall h \geq 1, \text{Cov}(x_t, x_{t+h})$ may depend on h but not on t .

Weak Dependence

Stationarity concerns the *joint distributions* of a process as you move in time. **Weak dependence** concerns *how strongly related* x_t and x_{t+h} are as they get further apart (h grows). We need

$$\text{Corr}(x_t, x_{t+h}) \rightarrow 0 \quad \text{quickly as } h \rightarrow \infty$$

to invoke the LLN and CLT and therefore justify the use of OLS.

Asymptotic OLS in time series

In a large sample, the asymptotic properties of OLS ensure consistency when TS1'–TS3' hold:

- TS1': linearity in parameters *and* weak dependence (and stationarity).
- TS3': contemporaneous exogeneity, $E[u_t | \mathbf{x}_t] = 0$.

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Exercises



Exercise 10.2: Setup

Let $\{e_t : t = -1, 0, 1, \dots\}$ be a sequence of i.i.d. random variables with mean zero and variance one. Define a stochastic process by:

$$x_t = e_t - 0.5 e_{t-1} + 0.5 e_{t-2}, \quad t = 1, 2, \dots$$

Sub-questions

skip to Exercise 10.3 →

- (a) Find $E(x_t)$ and $V(x_t)$. Do either depend on t ?
- (b) Show that $\text{Corr}(x_t, x_{t+1}) = -0.5$ and $\text{Corr}(x_t, x_{t+2}) = 1/3$.
- (c) What is $\text{Corr}(x_t, x_{t+h})$ for $h > 2$?
- (d) Is x_t an asymptotically uncorrelated process?

Exercise 10.2(a): Mean and Variance

Mean.

$$E(x_t) = E(e_t - 0.5 e_{t-1} + 0.5 e_{t-2}) = E(e_t) - 0.5 E(e_{t-1}) + 0.5 E(e_{t-2}) = 0.$$

Variance. Because e_t is i.i.d., $\text{Cov}(e_t, e_s) = 0$ for all $t \neq s$, so

$$V(x_t) = V(e_t) + 0.25 V(e_{t-1}) + 0.25 V(e_{t-2}) = 1 + 0.25 + 0.25 = \frac{3}{2}.$$

Conclusion

Neither $E(x_t)$ nor $V(x_t)$ depends on t .

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Exercise 10.2(b): First-Order Correlations

One-step covariance.

$$\begin{aligned}\text{Cov}(x_t, x_{t+1}) &= E[(e_t - 0.5 e_{t-1} + 0.5 e_{t-2})(e_{t+1} - 0.5 e_t + 0.5 e_{t-1})] \\ &= -0.5 E[e_t^2] - 0.5^2 E[e_{t-1}^2] = -0.5 - 0.25 = -\frac{3}{4}.\end{aligned}$$

All cross-products of e 's with different time indices vanish (i.i.d. with zero mean: $E[e_t e_s] = E[e_t]E[e_s] = 0$), and $E[e_t^2] = V(e_t) = 1$.

$$\text{Corr}(x_t, x_{t+1}) = \frac{\text{Cov}(x_t, x_{t+1})}{V(x_t)} = \frac{-3/4}{3/2} = -\frac{1}{2}.$$

Two-step covariance.

$$\text{Cov}(x_t, x_{t+2}) = E[(e_t - 0.5 e_{t-1} + 0.5 e_{t-2})(e_{t+2} - 0.5 e_{t+1} + 0.5 e_t)] = 0.5 E[e_t^2] = \frac{1}{2}.$$

$$\text{Corr}(x_t, x_{t+2}) = \frac{1/2}{3/2} = \frac{1}{3}.$$

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Exercise 10.2(c): Higher-Order Correlations

Consider $h = 3$:

$$\text{Cov}(x_t, x_{t+3}) = E[(e_t - 0.5 e_{t-1} + 0.5 e_{t-2})(e_{t+3} - 0.5 e_{t+2} + 0.5 e_{t+1})]$$

x_t depends only on e_{t+j} for $j \leq 0$, while x_{t+3} depends only on e_{t+j} for $j > 0$. The two sets of innovations do not overlap, and $E[e_t e_s] = 0$ for $t \neq s$.

Conclusion

$$\text{Corr}(x_t, x_{t+h}) = 0 \quad \forall h > 2.$$

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Exercise 10.2(d): Asymptotically Uncorrelated?

Definition. An *asymptotically uncorrelated* process is a covariance stationary process for which

$$\text{Cov}(x_t, x_{t+h}) \rightarrow 0 \quad \text{as } h \rightarrow \infty.$$

Answer: Yes

Terms more than two periods apart are *actually* uncorrelated (part c). It is therefore immediate that $\text{Corr}(x_t, x_{t+h}) \rightarrow 0$ as $h \rightarrow \infty$.

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Exercise 10.3: Setup

Suppose the time series process y_t is generated by

$$y_t = z + e_t, \quad t = 1, 2, \dots$$

where e_t is an i.i.d. sequence with mean zero and variance σ_e^2 . The random variable z does *not change over time*; it has mean zero and variance σ_z^2 . Each e_t is uncorrelated with z .

Sub-questions

skip to Exercise 10.5 →

- (a) Find $E(y_t)$ and $V(y_t)$. Do they depend on t ?
- (b) Find $\text{Cov}(y_t, y_{t+h})$. Is y_t covariance stationary?
- (c) Show that $\text{Corr}(y_t, y_{t+h}) = \sigma_z^2 / (\sigma_z^2 + \sigma_e^2)$.
- (d) Does y_t satisfy the intuitive requirement for being asymptotically uncorrelated?

Exercise 10.3(a): Mean and Variance

Mean.

$$E(y_t) = E(z + e_t) = E(z) + E(e_t) = 0.$$

Variance.

$$V(y_t) = V(z + e_t) = V(z) + V(e_t) + 2 \operatorname{Cov}(z, e_t) = \sigma_z^2 + \sigma_e^2 + 2 \times 0 = \sigma_z^2 + \sigma_e^2.$$

(Recall: z and e_t are uncorrelated by assumption.)

Conclusion

Neither $E(y_t)$ nor $V(y_t)$ depends on t .

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Exercise 10.3(b): Covariance Stationarity

For $h = 0$: $\text{Cov}(y_t, y_t) = V(y_t) = \sigma_z^2 + \sigma_e^2$. For $h > 0$:

$$\begin{aligned}\text{Cov}(y_t, y_{t+h}) &= E[y_t y_{t+h}] = E[(z + e_t)(z + e_{t+h})] \\ &= E(z^2) + E(z e_{t+h}) + E(e_t z) + E(e_t e_{t+h}) \\ &= E(z^2) = \sigma_z^2,\end{aligned}$$

because e_t is i.i.d. with zero mean and z is uncorrelated with every e_t .

Conclusion

$E(y_t)$ and $V(y_t)$ do not depend on t (part a), and $\text{Cov}(y_t, y_{t+h}) = \sigma_z^2$ depends neither on t nor on h (for $h > 0$). Therefore, y_t is **covariance stationary**.

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Exercise 10.3(c): The Correlation Function

Combining parts (a) and (b):

$$\text{Corr}(y_t, y_{t+h}) = \frac{\text{Cov}(y_t, y_{t+h})}{\text{sd}(y_t) \text{sd}(y_{t+h})} = \frac{\text{Cov}(y_t, y_{t+h})}{V(y_t)} = \frac{\sigma_z^2}{\sigma_z^2 + \sigma_e^2} > 0.$$

Observation

The correlation is the same positive constant for every $h > 0$ — it depends only on the variance ratio of the time-invariant component to the total variance.

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Exercise 10.3(d): Asymptotically Uncorrelated?

Answer: No

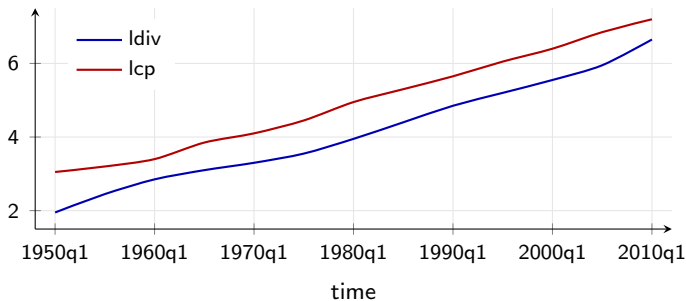
The correlation between y_t and y_{t+h} equals $\sigma_z^2/(\sigma_z^2 + \sigma_e^2)$ for every $h > 0$. Rather than approaching zero as h gets larger, the correlation is **always the same positive value**.

Why? The persistent correlation across time is driven by the presence of the time-constant component z , which never washes out no matter how far apart the two observations are.

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Exercise 10.5: Setup

Consider the dividends (\$bn) and corporate profits (\$bn) for the U.S. Let l_{div} and l_{cp} denote the logarithms of dividends and corporate profits respectively.



Exercise 10.5: Sub-questions

- (a) Comment on the pattern of both series and classify them with respect to stationarity.
- (b) Would you expect $\log(\text{div})_t = \beta_0 + \beta_1 \log(\text{cp})_t + u_t$ to be a spurious regression?
- (c) Relate the estimated results to your prediction.
- (d) Compare the trend coefficient and the differenced-model intercept.
- (e) Comment on the differenced series and the order of integration.
- (f) Test for serial correlation of order 1 in the residuals of both regressions.

Exercise 10.5(a): Pattern & Stationarity

Both series are non-stationary

The two series *trend together* over the sample. The mean is clearly increasing over time — a basic violation of covariance stationarity (constant mean).

Intuition. Stationarity means that the statistical properties of the data-generating process *do not change over time*: the way the time series varies should not itself vary across time. The upward drift in both *ldiv* and *lcp* rules this out.

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Exercise 10.5(b): Spurious Regression?

$$\log(\text{div})_t = \beta_0 + \beta_1 \log(\text{cp})_t + u_t$$

Yes — most likely a spurious regression

Both series trend in the same direction, and the trend is not accounted for. We would wrongly conclude that corporate profits have a positive effect on dividends merely because both share an upward drift.

Diagnosis: omitted-variable bias. The dependent variable is trending, so the time trend is captured in the error term. The explanatory variable $\log(\text{cp})_t$ is also trending, hence correlated with the omitted time trend. Classic OVB conditions hold.

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Exercise 10.5(c): Estimation Results

$$\widehat{\log(div)}_t = \underset{(-26.50)}{-1.33} + \underset{(109.63)}{1.13} \log(cp)_t \quad R^2 = 0.98$$

Hallmarks of a spurious regression

- Slope coefficient greater than one
- An extremely high t -ratio (109.63)
- An implausibly large $R^2 = 0.98$

All three are classic symptoms when two trending series are regressed on each other without controlling for the trend.

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Exercise 10.5(d): Trend vs. Differenced Intercept

Trend regression ($y_t = \alpha_0 + \alpha_1 t + \beta_1 x_t + u_t$):

$$\widehat{\log(div)}_t = \underset{(4.14)}{0.436} + \underset{(17.71)}{0.013} t + \underset{(10.55)}{0.425} \log(cp)_t \quad R^2 = 0.9914$$

Differenced regression ($\Delta y_t = \alpha_1 + \beta_1 \Delta x_t + \Delta u_t$):

$$\Delta \widehat{\log(div)}_t = \underset{(9.24)}{0.019} + \underset{(1.94)}{0.059} \Delta \log(cp)_t \quad R^2 = 0.0154$$

Interpretation

Differencing $y_t = \alpha_0 + \alpha_1 t + \beta_1 x_t + u_t$ yields $\Delta y_t = \alpha_1 + \beta_1 \Delta x_t + \Delta u_t$. The intercept of the differenced model *equals* the trend coefficient of the level model: both estimate the average per-period change in y_t when Δx_t is zero.

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Exercise 10.5(d): Numerical Comparison

Trend coefficient (level model). $\hat{\alpha}_1 = 0.013$. On average, ceteris paribus, the *quarterly percentage change* in dividends is 1.3%. **Intercept (differenced model).**

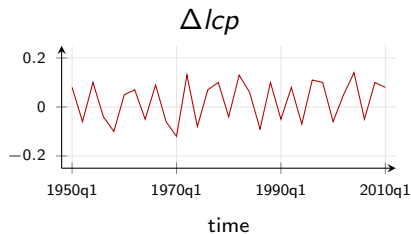
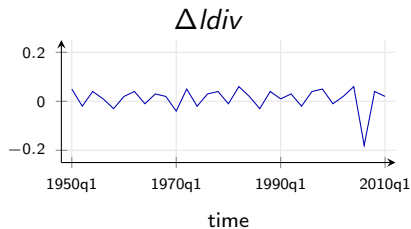
$\hat{\alpha}_1 = 0.019$. The expected value of $\Delta \log(\text{div})_t$ when $\Delta \log(\text{cp})_t = 0$ is 0.019, i.e. the average quarterly percentage change in dividends is 1.9%.

Why the slight discrepancy?

Both estimators *target* the same parameter (the average period-on-period change), but they exploit different variation in the data, so the two point estimates differ a bit numerically. Conceptually, however, they are estimating the same quantity.

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Exercise 10.5(e): Differenced Series



Order of integration

Both differenced series fluctuate around a roughly constant mean — they appear to be $I(0)$. This implies the original series $\log(div)_t$ and $\log(cp)_t$ were $I(1)$ (i.e. stationary in first differences but not in levels).

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Exercise 10.5(f): Serial Correlation in Residuals

Auxiliary regressions of OLS residuals on their own first lag:

$$\widehat{resid1}_t = 0.0008 + 0.9202 resid1_{t-1} \quad R^2 = 0.85$$

(0.16) (37.54)

$$\widehat{resid2}_t = -0.0001 - 0.0911 resid2_{t-1} \quad R^2 = 0.008$$

(-0.07) (-1.42)

Test for AR(1) errors with strictly exogenous regressors. (1) OLS-regress y_t on $x_1, \dots, x_k$ to get $\hat{u}_t$; (2) regress $\hat{u}_t$ on $\hat{u}_{t-1}$; (3) test $H_0 : \gamma_1 = 0$ vs. $H_a : \gamma_1 \neq 0$.

Conclusion

Levels: $t = 37.54$ — strong evidence of AR(1) (OLS still consistent, inference invalid). **Differences:** $|t| = 1.42$ — no evidence of AR(1). Differencing made the series $I(0)$ and resolved autocorrelation.

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